A PROJECT REPORT ON

CAMPUSAI: AN INTELLIGENT AI-POWERED COLLEGE ASSISTANT CHATBOT AND KNOWLEDGE BASE SYSTEM

A Project Report Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF BUSINESS ADMINISTRATION (COMPUTER APPLICATIONS)
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Academic Year: 2025 – 2026

CERTIFICATE OF APPROVAL

This is to certify that the project report entitled **"CampusAI: An Intelligent AI-Powered College Assistant Chatbot and Knowledge Base System"** submitted by **Kanishk Singh** (Seat No: BBA-CA-301) and **Akash Rana** (Seat No: BBA-CA-302) is a bonafide work carried out by them under my supervision and guidance, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Business Administration (Computer Applications) / BCA** during the academic year **2025–2026**.

To the best of my knowledge, the matter presented in this report has not been submitted to any other University or Institute for the award of any degree or diploma.

Prof. [Guide Name]
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STUDENT DECLARATION

We hereby declare that the project entitled **"CampusAI: An Intelligent AI-Powered College Assistant Chatbot and Knowledge Base System"** written and submitted by us under the guidance of **Prof. [Guide Name]** is our original work. The software architecture, algorithms, database schema, and documentation presented in this report have not been previously submitted to any other University or Institution.

Date: 25th October 2026
Place: Pune, Maharashtra

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ABSTRACT

In higher educational institutions, administrative offices are consistently overwhelmed with repetitive inquiries regarding admissions eligibility, courses, fees, examination timetables, and campus facilities. Traditional channels—such as physical inquiry counters and static website notices—suffer from working-hour constraints, prolonged queues, and poor mobile searchability.

To resolve these bottlenecks, this project presents **CampusAI**, a 24/7 intelligent conversational web application designed specifically for college administrative and student support ecosystems. Developed using **Python (Flask)**, **SQLite (3NF Normalized)**, **HTML5**, **CSS3**, and asynchronous **JavaScript (Fetch API)**, CampusAI delivers sub-second response times for student inquiries.

CampusAI introduces an innovative **Dual-Tier Hybrid Response Architecture**: Tier-1 uses deterministic text normalization, stop-word filtering, domain synonym expansion, and Jaccard similarity to match against verified FAQs; Tier-2 invokes a context-bounded Large Language Model (Google Gemini API) only when conversational flexibility is required. A strict **Zero-Hallucination Policy** ensures that the bot never invents dates, fees, or policies—politely routing students to the physical office if unverified. Additionally, a protected **Admin Dashboard** secured with salted PBKDF2:SHA256 password hashing enables real-time CRUD management of FAQs. The automated test suite achieved a 100% pass rate across all 23 unit and integration test cases.

Keywords: Artificial Intelligence, Natural Language Processing, Conversational Chatbot, Higher Education, Flask REST API, 3NF Database, Zero-Hallucination Policy.

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CHAPTER 1: INTRODUCTION

1.1 Project Overview & Background

In modern higher educational institutions, administrative offices are consistently inundated with inquiries regarding admission criteria, course structures, semester fees, examination timetables, library policies, and academic notices. Traditionally, students obtain this information by visiting physical administrative counters or navigating complex static websites with downloadable PDF circulars. These methods introduce high latency, manual overhead, and frustration during peak admission and exam periods. **CampusAI** provides an automated, 24/7 web-based assistant that immediately answers natural-language questions with verified academic accuracy.

1.2 Problem Statement

"To design, implement, and validate an automated, 24/7 responsive, web-based College Assistant Chatbot that accurately resolves student academic and administrative inquiries using a dual-tier hybrid NLP matching and LLM fallback engine, supported by a 3NF relational database and a secure administrative dashboard that prevents artificial intelligence hallucinations and allows non-technical administrators to manage institutional knowledge in real time."

1.3 Project Objectives

- **Sub-Second Instant Assistance:** Deliver sub-second (under 300ms) answers to student questions 24/7.
- **Dual-Tier Hybrid Query Evaluation:** Combine deterministic semantic token matching with context-bounded LLM fallback.
- **Zero-Hallucination Academic Safety:** Strictly prevent fabrication of fees, dates, or contact details.
- **Normalized Relational Database (3NF):** Implement clean SQLite schema for FAQs, administrators, and session logs.
- **Real-Time Admin Management Portal:** Provide secure CRUD controls with immediate reflection in chatbot sessions.
- **Searchable Knowledge Directory:** Interactive, categorized accordion directory with real-time keyword search.
- **Automated Test Verification:** 100% test coverage across 23 unit, security, and integration test cases.

1.4 Scope of the System

CampusAI covers six primary domains: Admissions, Courses, Fees, Examinations, Facilities, and General Policies. The system runs entirely locally with SQLite, connects optionally to Google Gemini for conversational fallback, and remains fully functional offline using local NLP rules.

CHAPTER 2: LITERATURE SURVEY & FEASIBILITY STUDY

2.1 Existing Systems vs. CampusAI

Prior to engineering CampusAI, existing institutional support systems were analyzed: physical office counters, static PDF circular websites, keyword-based rule bots, and unguarded LLM wrappers. The findings are summarized in Table 2.1.

Metric	Physical Counter	Static Website	Generic LLM	CampusAI (Ours)
Availability	10 AM - 4 PM	24/7 (Passive)	24/7	**24/7 (Active)**
Response Latency	15-45 mins	Manual Search	1.5-4.0 sec	**< 300 ms**
Hallucination Risk	None	None	High (Severe)	**Zero (Guarded)**
Admin Updation	Manual briefing	Webmaster file upload	Prompt engineering	**Instant Web GUI**
Offline Resilience	N/A	High	Fails completely	**High (Local NLP)**

2.2 Feasibility Analysis

- **Technical Feasibility:** Built using mature open-source technologies (Python, Flask, SQLite, Vanilla JS) requiring no exotic infrastructure.
- **Operational Feasibility:** Chat interface mirrors familiar apps like WhatsApp, requiring zero user training for students or administrators.
- **Economic Feasibility:** Zero software licensing fees; runs on free-tier hosting (Render.com) or local intranet servers.

CHAPTER 3: SYSTEM REQUIREMENTS SPECIFICATION (SRS)

3.1 Hardware and Software Specifications

Category	Component	Specification
Hardware	Client Device	Smartphone / PC (1 GHz CPU, 1 GB RAM, Modern Browser)
Hardware	Server Host	1 vCPU, 512 MB RAM, 500 MB Storage (Local or Cloud)
Software	Operating System	macOS, Linux (Ubuntu), or Windows 10/11
Software	Backend Engine	Python 3.10+ / Flask 3.1.3
Software	Database	SQLite 3 (ACID-compliant, embedded relational)
Software	Production WSGI	Gunicorn 21.0+ / 26.2+
Software	Frontend Stack	Semantic HTML5, CSS3 Variables, Asynchronous ES6 JS
Software	Security	Werkzeug PBKDF2:SHA256 Password Hashing
Software	AI Layer	Google GenAI SDK (gemini-2.5-flash / gemini-3.8-flash)

3.2 Functional Requirements Summary

1. **Student Chatbot Interface:** Natural-language input, typing indicator, message history, suggested question chips, enter-to-send.
2. **Searchable FAQ Directory:** Category filtering (Admissions, Courses, Fees, Exams, Facilities, General), accordion view, real-time search.
3. **Admin Authentication:** Protected login, salted PBKDF2 password verification, HTTP-only session cookies, unauthorized access redirects.
4. **Admin Management Portal:** Real-time CRUD operations for FAQs, telemetry metrics cards, live search, immediate database reflection.
5. **Safe AI Fallback:** Context-bounded LLM generation with strict refusal to fabricate unverified dates, fees, or policy rules.

CHAPTER 4: SYSTEM ANALYSIS & DESIGN

4.1 Relational Database Design (3NF Normalization)

The database schema was engineered strictly in Third Normal Form (3NF) to eliminate data redundancy and support instantaneous query lookups:

Table	Field	Type	Constraints	Description
admins	id	INTEGER	PK, Auto-Inc	Admin identifier
admins	username	VARCHAR(80)	NOT NULL, UNIQUE	Login username
admins	email	VARCHAR(120)	NOT NULL, UNIQUE	Admin email address
admins	password_hash	VARCHAR(255)	NOT NULL	PBKDF2:SHA256 salted hash
faqs	id	INTEGER	PK, Auto-Inc	FAQ identifier
faqs	question	TEXT	NOT NULL	Student query text
faqs	answer	TEXT	NOT NULL	Authoritative answer
faqs	category	VARCHAR(50)	NOT NULL, INDEX	Categorical domain
chat_messages	id	INTEGER	PK, Auto-Inc	Message identifier
chat_messages	session_id	VARCHAR(64)	NOT NULL, INDEX	UUID session token
chat_messages	role	VARCHAR(20)	NOT NULL	student / assistant
chat_messages	content	TEXT	NOT NULL	Message text body
chat_messages	source	VARCHAR(30)	DEFAULT 'database'	database / ai / fallback

4.2 Dual-Tier Hybrid NLP & Fallback Algorithm

When a student sends a message, CampusAI evaluates the query through a multi-stage decision pipeline:

1. **Text Normalization:** Lowercase, strip punctuation, remove English stop words.
2. **Domain Synonym Expansion:** Expand keywords (e.g., *'fee'* → *[tuition, cost, payment, charges]*).
3. **Jaccard Overlap Scoring:** Compute similarity $S = |Q \cap F| / |F|$ against all database FAQs.
4. **Threshold Check:** If $S \ge 0.45$, return authoritative database answer instantly (< 20ms).
5. **Bounded AI Fallback:** If $S < 0.45$ and LLM API is available, invoke Gemini with strict system context.
6. **Zero-Hallucination Guard:** If unverified, respond politely and provide physical college office contact details.

CHAPTER 5: SOFTWARE TESTING & VERIFICATION

5.1 Automated Test Suite Execution (23 Test Cases)

A rigorous automated test suite was constructed using Python's unittest framework. All 23 test cases executed successfully with a 100% pass rate in 1.185 seconds:

Test ID	Test Scenario	Route / Component	Result	Status
TC-01	Landing Page Load	GET /	HTTP 200 OK	**PASS**
TC-02	Student Chat Page Load	GET /chat	HTTP 200 OK	**PASS**
TC-03	FAQ Directory Page Load	GET /faqs	HTTP 200 OK	**PASS**
TC-04	Admin Login Page Load	GET /admin/login	HTTP 200 OK	**PASS**
TC-05	System Health Check API	GET /api/health	HTTP 200 healthy	**PASS**
TC-06	List All Seeded FAQs	GET /api/faqs	20 FAQs returned	**PASS**
TC-07	Filter FAQs by Category	GET /api/faqs?category=Fees	Fee FAQs only	**PASS**
TC-08	Search FAQs by Keyword	GET /api/faqs/search?q=fees	Matches returned	**PASS**
TC-09	Empty Search Result	GET /api/faqs/search?q=xyz	Empty array []	**PASS**
TC-10	Chatbot Empty Query Guard	POST /api/chat (empty)	HTTP 400 Bad Request	**PASS**
TC-11	Conversational Greeting	POST /api/chat ('hello')	Friendly greeting	**PASS**
TC-12	Admissions FAQ Match	POST /api/chat ('how to apply')	Admission FAQ	**PASS**
TC-13	Fee Structure FAQ Match	POST /api/chat ('what are fees')	Fee schedule FAQ	**PASS**
TC-14	Unknown Query Non-Hallucination	POST /api/chat ('alien')	Office referral	**PASS**
TC-15	AI Provider Failure Fallback	Simulated LLM Exception	Local FAQ fallback	**PASS**
TC-16	Real-Time FAQ Reflection	Create FAQ → Query	Immediate answer	**PASS**
TC-17	Admin Login Success	POST /api/admin/login	HTTP 200 OK Session	**PASS**
TC-18	Admin Wrong Password Reject	POST /api/admin/login	HTTP 401 Reject	**PASS**
TC-19	Admin Blank Fields Reject	POST /api/admin/login	HTTP 400 Bad Request	**PASS**
TC-20	Unauthenticated Dashboard Guard	GET /admin/dashboard	HTTP 302 Redirect	**PASS**
TC-21	Unauthenticated Admin API Guard	GET /api/admin/stats	HTTP 401 Reject	**PASS**
TC-22	Admin FAQ CRUD Lifecycle	POST/PUT/DELETE FAQ	Full CRUD Success	**PASS**
TC-23	Admin Session Logout	POST /api/admin/logout	Session Cleared	**PASS**

CHAPTER 6: CONCLUSION & FUTURE SCOPE

6.1 Conclusion

CampusAI demonstrates that academic institutions can achieve high-performance, 24/7 student inquiry resolution without expensive enterprise subscriptions or artificial intelligence hallucinations. By unifying deterministic local NLP matching with an authoritative 3NF relational database and a bounded generative AI fallback, the system guarantees factual integrity, sub-second response speeds, and frictionless administrative management.

6.2 Future Enhancements

- **Multilingual Processing:** Integration of regional Indian language NLP models (Marathi, Hindi).
- **Voice & Speech Interaction:** Hands-free voice inquiry support via Web Speech API and Text-to-Speech engines.
- **Omnichannel Connectors:** Direct integration with WhatsApp Business API and Telegram bots.
- **Student ERP Sync:** Secure student authentication for personalized fee balance and exam grade queries.

REFERENCES & BIBLIOGRAPHY

1. Grinberg, Miguel. (2018). *Flask Web Development: Developing Web Applications with Python*. O'Reilly Media.

2. Jurafsky, D., & Martin, J. H. (2023). *Speech and Language Processing*. Stanford University / Pearson.

3. Pallets Projects. (2024). *Flask Documentation (Version 3.1.x)*. Available at: https://flask.palletsprojects.com/

4. SQLAlchemy Authors. (2024). *SQLAlchemy 2.0 Documentation*. Available at: https://docs.sqlalchemy.org/

5. Google Cloud. (2024). *Google GenAI SDK & Gemini API Documentation*. Available at: https://ai.google.dev/

6. Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach*. McGraw-Hill.